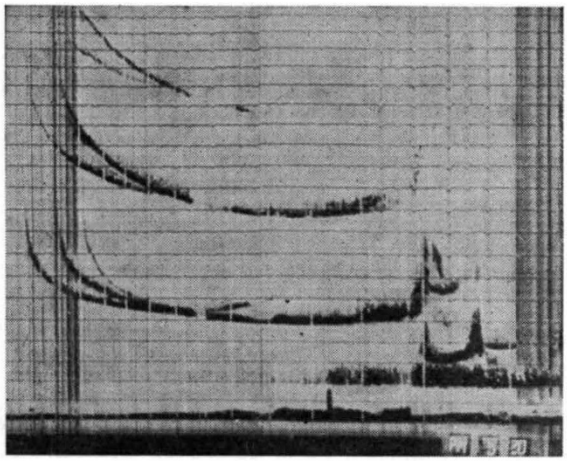
Triple Magneto Ionic Splitting of Rays Reflected from the F_2 Region

A RECORD for 1700 hr. local time on November 9, 1946, from the Hobart *P'f* recorder operating in the frequency range 1·5–13 Mc./s. is reproduced below.



The record shows the ordinary and extraordinary rays usually present from F_2 with critical frequencies of 7·6 and 8·5 Mc./s. respectively, as well as a third ray the critical frequency of which is 7·0 Mc./s. The critical frequencies of these rays correspond approximately to the Appleton–Hartree[1] reflexion condition $\mu = 0$ except that, neglecting collision, one would expect the third ray to have a critical frequency of 6·7 Mc./s. Discrepancies of about this magnitude and sign which are not ascribable to experimental error occur in all the 'triple split' measurements taken and may perhaps be accounted for if collision effects were included in the calculations. The possible existence of triple magneto ionic splitting in a layer where the variation of electron density with height is very rapid was first discussed by Mary Taylor[2], and the phenomenon has been observed by Toshniwal[3] and Liev Harang[4]; but the record shown is one of the best examples of it that we have seen.

Thirty-nine definite examples of this triple splitting have been observed in some 50,000 records spaced at regular intervals over a period of a year. Other records possibly showing this effect (quadruple and even higher multiple forking) have been discounted as examples of multiple magneto ionic splitting because of indications of stratification in F_2 or very spread echoes. The triple split records are generally accompanied by sporadic E ionization and seem more likely to occur under disturbed conditions. Triple splits have been observed at nearly every hour of the day, but they occur most frequently between the hours of 1700 and 2000. Occasionally they appear on the records over a period of several hours, starting or finishing as multiple forked records in the stratified layer. There appears on the Hobart data to be no evidence for the seasonal variation noticed by Harang. The figures for the number of occasions present (they may be either single occurrences or last for several hours) over twelve months are: 1946, September 2, October 4, November 3, December 0; 1947, January 0, February 3, March 5, April 1, May 0, June 2, July 5 and August 4. Further data on the occurrence of the phenomena will be analysed as they become available.

The records from the Hobart *P'f* recorder are measured by the University of Tasmania for the Radio Research Board of the Council of Scientific and Industrial Research, and I would like to acknowledge the help received from that body in this and other ionosphere investigations.

GORDON NEWSTEAD
Electrical Engineering Department,
University of Tasmania,
Hobart.
Nov. 5.

[1] Appleton, E. V., *J. Inst. Elect. Eng.*, **71**, 642 (1932).
[2] Taylor, Mary, *Proc. Phys. Soc.*, **45**, 245 (1933).
[3] Toshniwal, G. R., *Nature*, **155**, 471 (1935).
[4] Liev Harang, *Terr. Mag.*, **41**, 143 (1936).

Variable Source of Radio Frequency Radiation in the Constellation of Cygnus

COSMIC or galactic noise was discovered by Jansky[1] in 1931; but its exact origin has remained uncertain. It is generally supposed to originate from collisions in interstellar matter[2]; but there are divergencies between existing theory and experimental results, particularly at lower radio frequencies[3]. Hey, Parsons and Phillips[4] discovered variations in the intensity of galactic noise from the direction of the constellation of Cygnus, with a period of about one minute—suggesting that this particular radiation has its origin in a discrete source.

During the past three months, we have made a study of this region, mainly on 100 Mc./s., but also occasionally on 60, 85 and 200 Mc./s. The technique employed was to observe the region rising over the sea with aerials situated on a high cliff, as described by Pawsey, Payne-Scott and McCready[5]. Due to interference between the direct ray and the ray reflected from the sea, a lobe pattern is obtained which gives rise to a succession of maxima and minima. An estimate of the size of the source can be made from the relative heights of maxima and minima, and an accurate position found from the times of occurrence of minima.

Small aerial arrays were used—one or two Yagis—and considerable care was taken with receiver stabilization enabling receiver noise to be balanced out and the input signal amplified for presentation on a recording milliammeter. At 100 Mc./s. receiver fluctuations represented an input signal of less than 3×10^{-24} watts m.$^{-2}$(c./s.)$^{-1}$.

The general results of our investigations are:

(1) The variations from Cygnus originate in an area of less than 8′ angular width. This is an upper estimate and the source may well be effectively a point.

(2) The location of the source is R.A. 19 hr. 58 min. 47 sec. ± 10 sec. Dec. +41° 41′ ± 7′.

(3) The radiation consists of two components, one constant and the other showing considerable variation over short and long periods. In Fig. 1, a typical record of the source rising, the two components are separated by a dotted line. A record of the 'quiet' sun rising is shown for comparison in Fig. 2. The frequency in both cases was 100 Mc./s.; intensity scales are the same on both records.

(4) Observation on different frequencies shows that the intensity of the constant component has a shallow

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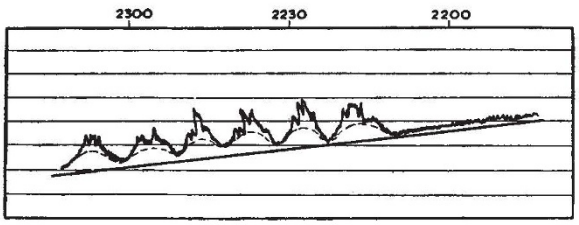


Fig. 1. Record of source in Cygnus rising

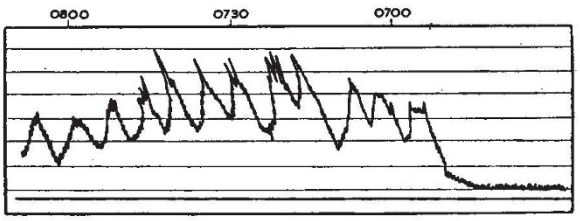


Fig. 2. Record of 'quiet' sun rising

peak in the region of 100 Mc./s., whereas that of the variable component increases rapidly with decreasing frequency. No variable component has so far been detected on 200 Mc./s.

(5) The periodicity of the variations decreases with decreasing frequency. No close correlation has been found between short-period variations on frequencies as close as 85 and 100 Mc./s., but the general activity of the variable component shows good correlation on different frequencies.

The determined position does not coincide with any outstanding stellar object, but is close to two seventh magnitude stars. It is notable that Hey[6] and Moxon[7] found a secondary maximum of galactic noise in Cygnus. The present evidence indicates that the noise from this region comes from a discrete source superimposed on a smooth continuum.

Taking the size of the sources as less than 8′, the constant component represents an effective temperature of more than 4×10^6 degrees Kelvin at 100 Mc./s., which makes a thermal origin of the noise doubtful. A mechanism similar to that proposed by Martyn[8] to account for enhanced solar radiation on metre wave-lengths (association of moving ionized matter and magnetic fields) may account for the phenomenon, though further evidence is required to throw more light on the problem.

Our thanks are due to Dr. J. L. Pawsey for his help and encouragement during this work, which is part of the programme of the Radiophysics Laboratory of the Council for Scientific and Industrial Research. Full details are being published elsewhere.

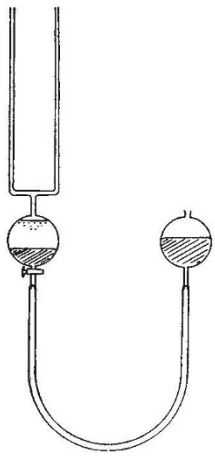
J. G. Bolton
G. J. Stanley

Division of Radiophysics,
Council for Scientific and Industrial Research,
Commonwealth of Australia.
Dec. 4.

[1] *Proc. Inst. Radio Eng.*, **20**, 1920 (1932).
[2] *Astrophys. J.*, **91**, 625 (1940).
[3] *Astrophys. J.*, **105**, 235 (1947).
[4] *Nature*, **158**, 234 (1946).
[5] *Nature*, **157**, 158 (1946).
[6] *Nature*, **157**, 297 (1946).
[7] *Nature*, **158**, 758 (1946).
[8] *Nature*, **159**, 26 (1947).

Particle-size Measurement

I have recently had occasion to devise a scheme for measuring the mean diameters of the particles of a powder. The method finally evolved has not been discovered during an examination of the available literature, and in view of its simplicity seems to merit description.



The essential features of the apparatus are shown in the diagram, the vertical limbs of the U-tube being in this case about 0·1 sq. cm. in area. A non-polar 'wetting' liquid such as carbon tetrachloride is floated on mercury in the reservoir beneath the U-tube. By raising the other reservoir, mercury is driven over, and the wetting liquid rises in the vertical tubes until the plug of powder is reached. The powder is held towards the top of one limb, by means of a plug of gauze or fibrous material of surface area less than that of the substance under study. When the powder is thoroughly wetted, the liquid is allowed to flow slowly back into the reservoir, control being effected by the tap.

The surface of the liquid is supported in one limb by the powder and falls in the other to a critical position at which the upper meniscus becomes detached from the plug. The difference in levels of the menisci (h) at the critical point is found to be connected with the mean radius (r) of the particles by the expression:

$$r = \frac{3mT}{hgad_1d_2l},$$

where m represents the weight of the powder used, T the surface tension of the wetting liquid, d_1 and d_2 the densities of liquid and powder respectively, while a and l represent the area of the tube and the length of the column of powder.

The value of l is found by flooding the column with liquid and, when free of air bubbles, sucking back on to the supporting plug.

It will be apparent that the use of a non-polar liquid is advocated in order to avoid adsorption on the powder, with the consequent increase in effective surface tension.

The accuracy of the method has been tested on a powder known to consist of nearly spherical particles, by comparing the results obtained by microscopical count with those found using carbon tetrachloride as wetting liquid. The mean result in each case was $r = 0{\cdot}017$ mm., the mean error of an observation of r by the method described above being 0·0008 mm.

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